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ORIGINAL PAPER

Posterior uveal melanoma in 321 Asian Indian patients: analysis based on the 8th edition of American Joint Committee Cancer classification

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Abstract
Purpose To evaluate the presenting features, treatment, and outcomes of posterior uveal melanoma (PUM) in Asian Indians based on the 8th edition of American Joint Committee on Cancer (AJCC) classification.
Methods Retrospective interventional case series of 321 Asian Indian patients with PUM.
Results Based on AJCC, PUM was classified into categories T1 ($n = 36$; 11%), T2 ($n = 74$; 23%), T3 ($n = 126$; 39%), and T4 ($n = 85$; 27%). Regarding tumor features, T4 was more likely to have pre-equatorial epicenter (vs T1 and T2; $p \leq 0.011$), iris abnormalities (vs T2 and T3; $p \leq 0.002$), and extraocular tumor extension (vs T3; $p = 0.001$), whereas T1 was more likely to have macular epicenter (vs T2, T3, T4; $p \leq 0.013$), lipofuscin deposits (vs T3 and T4; $p \leq 0.008$), and amelanotic tumors (vs. T4; $p = 0.003$). On multivariate analysis, factors predictive of systemic metastasis were increasing tumor thickness ($p = 0.002$) and extraocular tumor extension ($p = 0.009$). The 5-, 10-, and 15-year melanoma-related metastases rates were 0%, 0%, and 0% in T1, 0%, 60%, and 60% in T2, 7%, 40%, and 70% in T3 and 13%, 36%, and 76% in T4, respectively. Risk for metastasis was 1.23 times more for every 1-mm increase in tumor thickness and 9 times more with extraocular tumor extension.
Conclusion The AJCC 8th edition provides prognostic classification for PUM in Asian Indian patients. The significant risk factors for metastasis were increasing tumor thickness and extraocular tumor extension.

Keywords Eye · Tumor · Uvea · Choroid · Melanoma · Choroidal melanoma · AJCC 8th edition

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Introduction

Cancer staging is the scientific medical language defining the local, regional, and systemic extent of a cancer at a specified time. The American Joint Committee on Cancer (AJCC) established a way of standardized nomenclature called TNM where T is the primary tumor, N the lymph node extension, and M the distant metastasis for readily communicating the state of a cancer and to aid in treatment decision making and judging prognosis of cancer. AJCC was established in 1959 and published its first manual for cancer staging in 1977. Since then, a collaborative effort by the AJCC and the Union for International Cancer Control (UICC) provides the most comprehensive cancer anatomic staging data [1].

The original AJCC cancer staging manuals for uveal melanoma were based on previous epidemiological studies [1–3]. In 2003, the 6th edition was introduced which incorporated the Collaborative Ocular Melanoma Study definitions for posterior uveal melanoma (PUM) into the staging system. However, this staging did not assign any importance to ciliary body involvement. Moreover, it was neither evidence based nor validated by a cancer database [1, 2, 4]. Due to the efforts of 47 ocular oncologists from 11 countries, the first widely accepted staging system for eye cancer, the 7th edition of AJCC cancer staging manual, came into existence in 2009 [5–9] This T staging is mainly based on tabulated combinations of tumor thickness and largest basal dimension. It also takes into consideration ciliary body involvement and extrascleral tumor extension which are the two most important predictors for survival in uveal melanoma established in ophthalmic literature [1, 2]. The 8th edition of AJCC cancer staging manual was released in 2017 and has many similarities with the 7th edition classification for PUM (ciliary body and choroidal melanoma) with regard to T category.

Uveal melanoma is a deadly cancer arising from the stromal melanocytes of neural crest origin. Every year worldwide, 8000 new cases of uveal melanoma are reported predominantly affecting Caucasians and Hispanic population, while only a minority of Asians and Africans are affected [1, 10]. This most common intraocular malignancy in adults can affect iris, ciliary body, or choroid, of which choroidal melanoma is the most common. In a series of 8033 patients with uveal melanoma, the tumor was located in posterior uvea (ciliary body and choroid) in 96% cases [11].

The 7th AJCC cancer staging classification for PUM is evidence based and has been validated on a large cohort of White patients [5–9]. Though there are a few case series on choroidal melanoma in Asian Indian patients [12–16], the predictive value of AJCC 7th or 8th edition classification for PUM metastasis in Asian Indians has not been explored in ophthalmic literature. In this study, the authors aimed to evaluate the clinical presentation, treatment patterns, and outcomes (local recurrence and melanoma-related metastasis) of PUM in a large cohort of 321 Indian patients based on the tumor size category of AJCC 8th edition classification.

Methods

This was a retrospective, non-randomized, non-comparative, interventional case series. The clinical records of all patients with diagnosis of PUM managed at the Operation Eyesight Universal Institute for Eye Cancer, L V Prasad Eye Institute, Hyderabad, India, from January 1994 to May 2018 were reviewed. The patients with complete data documentation at first visit related to clinical presentation and tumor staging were included in the study, and those with incomplete data were excluded. Those with a minimum follow-up of 3 months or more were included for outcome analysis, and those with follow-up of < 3 months with unknown outcome or with any other concomitant cancers were excluded. This study adhered to the tenets of the Declaration of Helsinki and was approved by the Institutional Ethics Committee, L V Prasad Eye Institute, Hyderabad, India.

The patient data were reviewed retrospectively for demographic features, clinical features, management, and outcomes (local and systemic). The recorded demographic data included age at diagnosis, gender, and occupation. Presenting symptoms and their duration were also recorded. The following clinical features were recorded: visual acuity, intraocular pressure (mm Hg), anterior segment details based on slit lamp biomicroscopy and posterior segment details based on indirect ophthalmoscopic examination. The tumor data included tumor epicenter location (macula, between macula and equator, equator, pre-equatorial), quadrantic location of the tumor (superior, nasal, inferior, temporal, diffuse), greatest tumor thickness (in mm), largest basal diameter (in mm), color (melanotic, amelanotic, mixed), subretinal fluid, lipofuscin, distance of posterior tumor margin from optic disk, halo, drusen, ciliary body involvement, and extraocular extension (with largest diameter in mm). Tumor dimensions were objectively measured by ultrasonography. All findings were documented with color-coded fundus drawings, fundus photography, and ultrasound B-scan.

On the basis of tumor features (tumor basal diameter, thickness, location, and extraocular tumor extension), each tumor was categorized into T1, T2, T3, and T4 according to the 8th edition of the AJCC system. The details of primary treatment (transpupillary thermotherapy, Ruthenium-106 plaque brachytherapy, enucleation, orbital exenteration) were recorded. Systemic monitoring for metastasis was performed by a medical physician or oncologist with twice yearly physical examination, liver function tests, and abdominal and chest imaging (chest radiography or computed tomography or magnetic resonance imaging). The outcome regarding local tumor recurrence, systemic metastasis, and condition at last follow-up were recorded.

Statistical analysis

The statistical analysis was performed using the software Origin v7.0 (OriginLab Corporation, Northampton, MA, USA) and STATA v11.0 (StataCorp, College Station, TX, USA). The continuous data were checked for the normality of distribution by Shapiro–Wilk test and for the equality of variance by Levene test. The comparisons among stages T1 to T4 were performed by analysis of variance for continuous parametric data with equal variance, Kruskal–Wallis test for continuous nonparametric data or continuous parametric data with unequal variance and Chi-square test for categorical data. Multiple pair-wise comparisons between stages were performed by *t* test for continuous parametric data with equal variance, Mann–Whitney test for continuous nonparametric data or continuous parametric data with unequal variance and Chi-square test or Fisher exact test for categorical data. Univariate and multivariate logistic regression analyses were performed to evaluate the risk factors for metastasis and estimate odds ratio (OR) with 95% confidence interval (CI). Kaplan–Meier analysis was performed to estimate the rates of distant metastasis over time, and Cox proportional hazards regression model was used to compare survival rates between groups. A p-value of < 0.05 was considered statistically significant. For multiple pair-wise comparisons, Bonferroni correction was applied and a p-value < 0.017 was considered statistically significant.

Results

Based on the 8th edition of AJCC, PUM (n = 321) was classified into categories T1 (n = 36; 11%), T2 (n = 74; 23%), T3 (n = 126; 39%), and T4 (n = 85;

Table 1 Demographics of patients presenting with posterior uveal melanoma

Feature	All cases n = 32 n (%)	T1 n = 36 n (%)	T2 n = 74 n (%)	T3 n = 126 n (%)	T4 n = 85 n (%)	p-value
Age at presentation (years) Mean (median, range)	47 (46, 8 to 88)	49 (51, 10 to 76)	46 (46, 16 to 76)	46 (45, 12 to 76)	47 (47, 8 to 88)	0.69
Gender						
Male	186 (58)	20 (55)	39 (53)	74 (59)	53 (62)	0.93
Female	135 (42)	16 (44)	35 (47)	52 (41)	32 (38)	0.88
Occupation						
Outdoor	85 (26)	8 (22)	16 (22)	35 (28)	26 (31)	0.75
Indoor	236 (74)	28 (78)	58 (78)	91 (72)	59 (69)	0.96
Duration of symptoms (months) Mean (median, range)	6 (2, 0.1 to 104)	7 (4, 0.1 to 48)	6 (3, 0.1 to 96)	5 (2, 0.2 to 99)	8 (2, 0.1 to 104)	0.24
Symptoms						
Diminution of vision	250 (78)	30 (83)	57 (77)	95 (75)	68 (80)	0.98
Pain	21 (7)	1 (3)	1 (1)	12 (10)	7 (8)	0.13
Others	50 (15)	5 (14)	16 (19)	19 (15)	10 (12)	0.53
Follow-up duration (months) Mean (Rmedian, range)	27 (8, < 1 to 243)	32 (13, < 1 to 185)	26 (9, < 1 to 243)	24 (6, < 1 to 144)	30 (7, < 1 to 178)	0.61

27%). Demographic features are listed in Table 1. There was no significant difference in patient age, gender, duration of symptoms, presenting symptoms and follow-up duration among tumor categories.

The clinical features based on T category are listed in Table 2. The following features were more common in T4 tumors: preequatorial epicenter (vs. T1 and T2; $p \leq 0.011$), iris abnormalities (iris neovascularization, ectropion uveae and/or heterochromia iridis) (vs.T2 and T3; $p \leq 0.002$), and extraocular tumor extension (vs. T3; $p = 0.001$). The features more common in T1 tumors included macular epicenter (vs. T2, T3, T4; $p \leq 0.013$), lipofuscin deposits (vs T3 and T4; $p \leq 0.008$), and amelanotic tumors (vs. T4; $p = 0.003$). Ciliary body (CB) involvement was less common in T1 and T2 (vs. T3 and T4; $p \leq 0.001$), and secondary glaucoma was less common in T2 (vs. T3 and T4; $p \leq 0.013$). Diffuse tumor location was more common with T4 tumors ($p \leq 0.001$) compared to T2 and T3. There was no difference in quadrantic tumor location, distance from optic disk, presence of subretinal fluid and drusens among the different categories.

The ultrasonographic features are listed in Table 3. The most significant differences based on ultrasonographic tumor characteristics included tumor thickness ($p \leq 0.005$) and tumor basal dimension ($p < 0.0001$) which increased significantly with increasing T staging from T1 to T4.

The primary treatment modalities according to the tumor subcategory are listed in Table 4. Transpupillary thermotherapy (TTT) was more commonly performed for T1 (vs. T3 and T4; $p < 0.0001$) and T2 (vs. T3; $p = 0.002$) tumors. Among T2, T3, and T4 tumors, TTT was used as the primary treatment modality when the tumor thickness was < 4 mm. Of the 6 patients with T3 and T4 tumors who underwent TTT, globe salvage was achieved in 3 (50%) eyes, while the remaining 3 eyes required enucleation due to persistent tumor. Plaque brachytherapy was a more common treatment modality for T1 ($p < 0.0001$) and T2 ($p \leq 0.001$) (vs. T3 and T4) tumors. Enucleation was commonly required for T3 (vs T2; $p = 0.007$) and T4 (vs T1; $p \leq 0.002$) tumors. Orbital exenteration was a more common treatment modality for T4 tumors (vs T2 and T3; $p \leq 0.015$).

The outcome measures of PUM over a mean follow-up duration of 27 months (T1 = 32, T2 = 26, T3 = 24, T4 = 30 months, respectively) are listed in Table 5. Metastasis in T4 was significantly higher than T1 ($p = 0.014$) and T2 ($p = 0.006$). Mean time interval between initial diagnosis of PUM and systemic metastasis was 4.3 years and was not significantly different among different subgroups. The rates of local tumor recurrence were comparable among T categories.

Univariate and multivariate analyses of factors predicting systemic metastasis are listed in Table 6. On multivariate analysis, factors predictive of systemic metastasis were tumor thickness [OR = 1.23 (95% CI, 1.08–1.41); $p = 0.002$], and extraocular tumor extension [OR = 8.89 (95% CI, 1.73–45.53), $p = 0.009$].

On Kaplan–Meier (KM) time-to-event analysis of the outcome (Table 7), the 3-, 5-, and 10-year melanoma-related metastasis rates were 0%, 0%, and 0% for T1; 0%, 0%, and 60% for T2; 3%, 7%, and 40% for T3; and 13%, 13%, and 36% for T4 tumors, respectively. The KM estimate for 3-, 5-, 10-year survival rates was 100%, 100%, and 100% for T1; 100%, 100%, and 30% for T2; 98%, 98%, and 73% for T3; and 92%, 92%, and 67% for T4 tumors.

Discussion

AJCC cancer staging system is a valuable tool for patient counseling regarding metastasis-free survival to PUM patients. Owing to the increased occurrence of this tumor in Caucasians and Hispanics, most of the validation of the staging system has been performed on the same population. There exists a dearth of existing literature on validity of this prognostic tool in Asian Indians where the disease is rare. The TNM7 formulated a revamped classification system taking into account all the drawbacks of TNM6, which has been continued as the TNM8 with minimal changes in the N subcategory. In 2013, the European Ophthalmic Oncology Group performed a multicenter study including 7369 PUM and validated the TNM7, and showed that TNM7 is useful in prognostication of PUM [6]. In our study, we explored the clinical features, treatment, and treatment outcomes of PUM based on the AJCC 8th edition classification. Based on our study, T4 tumors are more likely to be located in the preequatorial region, and associated with iris abnormalities and extraocular tumor extension. T1 tumors are more likely to be located in the macula,

Table 2 Clinical features of patients presenting with posterior uveal melanoma

Feature	All cases n = 321 n (%)	T1 n = 36 n (%)	T2 n = 74 n (%)	T3 n = 126 n (%)	T4 n = 85 n (%)	p-value
Associated ocular(dermal) melanocytosis	15 (5)	3 (8)	3 (4)	3 (2)	6 (7)	0.34
Quadrantic tumor location						
Superior	54 (17)	4 (11)	11 (15)	29 (23)	10 (12)	0.23
Nasal	55 (17)	2 (6)	14 (19)	26 (21)	13 (15)	0.30
Inferior	61 (20)	8 (22)	20 (27)	22 (18)	11 (13)	0.29
Temporal	110 (34)	16 (44)	29 (39)	48 (38)	17 (20)	0.13
Diffuse	40 (12)	6 (17)	0 (0)	0 (0)	34 (40)	**< 0.0001**[a]
Anteroposterior tumor epicenter						
Macula	70 (22)	22 (61)	18 (25)	23 (18)	7 (8)	**< 0.0001**[b]
Macula to equator	134 (42)	8 (22)	41 (56)	49 (39)	36 (42)	0.17
Equator	37 (12)	3 (9)	7 (9)	20 (16)	7 (8)	0.38
Pre-equatorial	77 (24)	3 (9)	8 (10)	34 (27)	32 (38)	**0.004**[c]
Tumor color						
Pigmented	290 (91)	28 (78)	69 (94)	111 (89)	82 (97)	0.90
Amelanotic	24 (7)	7 (19)	4 (5)	11 (8)	2 (2)	**0.03**[d]
Mixed	7 (2)	1 (3)	1 (1)	4 (3)	1 (1)	0.75
Posterior tumor margin						
$\leq$ 3 mm from optic disk	152 (47)	24 (67)	38 (51)	56 (44)	34 (40)	0.44
$>$ 3 mm from optic disk	169 (53)	12 (33)	36 (49)	70 (56)	51 (60)	0.43
Associated features						
AC inflammation	5 (2)	0 (0)	2 (3)	3 (2)	0 (0)	0.39
Hyphema	4 (1)	1 (3)	2 (3)	1 (1)	0 (0)	0.37
NVI	18 (6)	1 (3)	1 (1)	8 (6)	8 (9)	0.17
Ectropion uveae and/or iris heterochromia	25 (8)	1 (3)	1 (1)	6 (5)	17 (20)	**< 0.0001**[e]
Secondary glaucoma	40 (12)	2 (6)	1 (1)	15 (12)	22 (26)	**< 0.0001**[f]
Subretinal fluid	287 (89)	31 (86)	66 (89)	106 (84)	84 (99)	0.88
Drusen	38 (12)	3 (8)	14 (19)	16 (13)	5 (6)	0.14
Lipofuscin	113 (35)	24 (67)	32 (43)	36 (29)	21 (25)	**0.02** [g]
Ciliary body involvement	98 (30)	0 (0)	7 (9)	45 (36)	46 (54)	**< 0.0001** [h]
Extraocular tumor extension	16 (5)	0 (0)	2 (3)	2 (2)	12 (13)	**< 0.0001**[i]

[a]Multiple pair-wise comparisons showed that T2 was significantly different from T1 ($p = 0.001$) and T4 ($p < 0.0001$), and T3 was significantly different from T1 (p < 0.0001) and T4 ($p < 0.0001$)

[b]Multiple pair-wise comparisons showed that only T1 was significantly different from T2 (p = 0.013), T3 ($p < 0.0001$) and T4 ($p < 0.0001$)

[c]Multiple pair-wise comparisons showed that only T4 was significantly different from T1 ($p = 0.011$) and T2 ($p = 0.002$)

[d]Multiple pair-wise comparisons showed that only T4 was significantly different from T1 ($p = 0.003$)

[e]Multiple pair-wise comparisons showed that only T4 was significantly different from T2 ($p = 0.001$) and T3 ($p = 0.002$)

[f]Multiple pair-wise comparisons showed that only T2 was significantly different from T3 ($p = 0.013$) and T4 ($p < 0.0001$)

[g]Multiple pair-wise comparisons showed that only T1 was significantly different from T3 ($p = 0.008$) and T4 ($p = 0.005$)

[h]Multiple pair-wise comparisons showed that T1 was significantly different from T3 ($p = 0.001$) and T4 ($p < 0.0001$), and T2 was significantly different from T3 ($p = 0.001$) and T4 ($p < 0.0001$)

[i]Multiple pair-wise comparisons showed that only T4 was significantly different from T3 ($p = 0.001$)

Table 3 Ultrasonographic features of patients presenting with posterior uveal melanoma

Feature	All cases n = 321 n (%)	T1 n = 36 n (%)	T2 n = 74 n (%)	T3 n = 126 n (%)	T4 n = 85 n (%)	p-value
Tumor basal diameter (mm) Mean (median, range)	13.3 (12.9, 3 to 30)	7.9 (8, 3 to 11)	11 (11, 5.6 to 17)	14 (13.9, 8 to 18)	19 (19, 11.4 to 30)	**<0.0001[a]**
Tumor thickness (mm) Mean (median, range)	8.6 (8.6, 2 to 19)	3.6 (3.5, 2 to 6)	6.4 (6, 3 to 12)	10 (10, 4 to 15)	11.8 (11.6, 2 to 19)	**<0.0001[b]**
Acoustic hollowness at tumor base	297 (93)	27 (75)	65 (88)	110 (87)	80 (94)	0.90
Choroidal excavation	279 (87)	30 (83)	59 (80)	107 (85)	70 (82)	0.99
Posterior shadowing	180 (56)	14 (39)	37 (50)	67 (53)	51 (60)	0.67

[a]Post hoc analysis showed that all pair-wise comparisons were significantly different from each other (all $p < 0.0001$)

[b]Post hoc analysis showed that all pair-wise comparisons were significantly different from each other (all $p < 0.0001$ except T3 vs T4 where $p = 0.005$)

associated with lipofuscin deposits, and be amelanotic. The metastatic rate in patients with T4 (18%) tumors was significantly more than those with T1 (0%) and T2 (3%) tumors, and was statistically significant. The most important factors predicting metastasis were increasing tumor thickness and extraocular tumor extension.

In a multicenter study of 3217 patients with PUM done by the AJCC Ophthalmic Oncology Task Force in 2015, 35% belonged to T1, 35% to T2, 24% to T3, and only 6% to T4 [5]. Similarly, in a study conducted at the Wills Eye Institute including 7731 patients with posterior uveal melanoma, majority of the patients belonged to T1 (46%) and T2 (27%) subgroups. The group-wise distribution in these studies performed at Europe and USA, respectively, is in sharp contrast to our study where majority of the patients belonged to T3 (39%) and a considerable subset of patients to T4 (26%) subcategory (Table 8) [7]. This difference is probably attributed to the fact that Asian Indian patients present late with advanced disease mostly because of lack of awareness about the disease. This fact is further corroborated by the Shields study, wherein non-Caucasian patients with PUM showed higher frequency of T4 tumors, similar to our study [7, 8]. The mean age of presentation of PUM was 47 years in our study against 58 years in Caucasians [7]. This finding suggests that Asian Indian patients present one decade earlier than the Caucasian patients [7]. Irrespective of age at presentation with PUM, both Caucasian and Asian Indian patients had a male preponderance for PUM. In our study, macula to equator was the most common location for tumor epicenter similar to the Caucasians; however, our population had significant percentage of preequatorial tumor epicenter which was negligible in their population. Mixed melanotic-amelanotic PUM contributed only 2% tumors in our population, whereas it was 31% in the Shields' study [7, 8].

Kaplan–Meier distant metastasis estimates at 5 years were 8%, 14%, 31%, 51% in the Shields' study, and 3%, 15%, 23%, 39% in the AJCC Ophthalmic Oncology Task Force worldwide multicenter study in the T1, T2, T3, T4 subcategories, respectively [5, 7]. In our study, 5-year Kaplan–Meier distant metastasis estimates were low at 0%, 0%, 7%, 13% in the T1, T2, T3, T4 subcategories, respectively. This could be related to racial differences in the outcomes. Similarly, the survival rates in our population (100%, 100%, 98%, 92%) were also higher than the Caucasians (96%, 92%, 81%, 70%) [7].

Compared to all studies in Caucasian population (Table 8) [5–7, 17], our study shows that despite late presentation with advanced disease, the distant metastatic rates are lower in all the T subgroups. This finding is also in contrast to outcomes in patients with primary cutaneous melanoma, where it has been noted that non-Caucasians are more prone to present with advanced disease and a higher disease-specific mortality compared to Whites [18]. In a study of 8100 patients with uveal melanoma including Caucasians (98%), Hispanics (1%), Asians, and African-Americans (< 1%), it was noted that the prognosis was similar [19]. The reason for lower rate of systemic metastasis in Asian Indian PUM patients may be attributed to younger age of presentation, shorter

Table 4 Primary treatment details of posterior uveal melanoma

Feature	All cases n = 321 n (%)	T1 *n* = 36 n (%)	T2 *n* = 74 n (%)	T3* *n* = 126 n (%)	T4 *n* = 85 n (%)	*p*-value
TTT	29 (8)	12 (34)	11 (15)	3 (2)	3 (3)	**<0.0001[a]**
Plaque radiotherapy	44 (14)	15 (41)	19 (26)	8 (6)	2 (2)	**<0.0001[b]**
Enucleation	174 (55)	5 (14)	26 (35)	90 (72)	53 (63)	**<0.0001[c]**
Orbital exenteration	8 (2)	0 (0)	0 (0)	1 (1)	7 (8)	**0.002[d]**
Denial of treatment	66 (21)	4 (11)	18 (24)	24 (19)	20 (24)	0.53

[a]Multiple pair-wise comparisons showed that T1 was significantly different from T3 ($p < 0.0001$) and T4 ($p < 0.0001$), and T2 was significantly different from T3 ($p = 0.002$)

[b]Multiple pair-wise comparisons showed that T1 was significantly different from T3 ($p < 0.0001$) and T4 ($p < 0.0001$), and T2 was significantly different from T3 ($p = 0.001$) and T4 ($p < 0.0001$)

[c]Multiple pair-wise comparisons showed that T1 was significantly different from T3 ($p < 0.0001$) and T4 ($p = 0.002$), and T2 was significantly different from T3 ($p = 0.007$)

[d]Multiple pair-wise comparisons showed that only T4 was significantly different from T2 ($p = 0.015$) and T3 ($p = 0.008$)

Table 5 Posterior uveal melanoma in Asian Indians: Outcomes

Feature	All cases n = 321 *n* (%)	T1 *n* = 36 n (%)	T2 n = 74 *n* (%)	T3 n = 126 *n* (%)	T4 n = 85 *n* (%)	*p*-value
Tumor recurrence	12 (3)	2 (6)	4 (5)	4 (3)	2 (2)	0.72
Metastasis	29 (9)	0 (0)	2 (3)	12 (10)	15 (18)	**0.006[a]**
Time interval between initial diagnosis of uveal melanoma and systemic metastasis (years) Mean (median, range)	4.3 (4.8, < 1 to 12.7)	N/A	8.2 (8.2, 7.1 to 9.2)	4.4 (5.4, < 1 to 11.8)	3.6 (0.9, < 1 to 12.7)	0.91[b]
Metastasis-related death	16 (5)	0 (0)	2 (3)	7 (6)	7 (8)	0.23

[a]Multiple pair-wise comparisons showed that only T4 was significantly different from T1 ($p = 0.014$) and T2 ($p = 0.006$)

[b]Comparison was done only between T3 and T4 because of small sample size in T1 and T2

Table 6 Factors related to metastasis in Asian Indians with posterior uveal melanoma: Univariate and multivariate logistic regression analyses

Feature	*p*-value	Odds ratio	95% confidence interval
Univariate analysis			
Tumor thickness (mm)	0.001	1.23	1.09–1.39
Largest basal diameter (mm)	0.01	1.14	1.03–1.26
AJCC T4 tumor category	0.009	2.92	1.31–6.50
Extraocular tumor extension	<0.0001	8.00	2.65–24.15
Multivariate analysis			
Increasing tumor thickness (mm)	0.002	1.23	1.08–1.41
Extraocular tumor extension	0.009	8.89	1.73–45.53

AJCC 8th edition of the American Joint Committee on Cancer classification

Table 7 Kaplan–Meier analysis of the outcome

Kaplan-Meier analysis	All cases $n = 321$	T1 $n = 36$	T2 $n = 74$	T3 $n = 126$	T4 $n = 85$	p-value
Distant metastasis						
1 year	4.4% ± 1.6%	0%	0%	2.7% ± 2%	13% ± 5.1%	**0.003**[a]
3 years	4.4% ± 1.6%	0%	0%	2.7% ± 2%	13% ± 5.1%	
5 years	5.9% ± 2.1%	0%	0%	6.8% ± 4.4%	13% ± 5.1%	
10 years	36.3% ± 8.4%	0%	60% ± 29.7%	40.2% ± 13.3%	36.4% ± 10.9%	
15 years	61.2% ± 12.6%	0%	60% ± 29.7%	70.1% ± 22.2%	76.2% ± 18.7%	
Survival						
1 year	97.5% ± 1.3%	100%	100%	98.3% ± 1.7%	92.3% ± 4.3%	0.14
3 years	97.5% ± 1.3%	100%	100%	98.3% ± 1.7%	92.3% ± 4.3%	
5 years	97.5% ± 1.3%	100%	100%	98.3% ± 1.7%	92.3% ± 4.3%	
10 years	65.4% ± 9.2%	100%	27.8% ± 23.2%	72.5% ± 12.2%	67.1% ± 13%	
15 years	44.8% ± 13.7%	100%	27.8% ± 23.2%	36.2% ± 26.3%*	33.6% ± 24.6%	

*Data available until 12 years

[a]Multiple pair-wise comparisons showed that none of the comparisons were statistically significant at $p < 0.017$ after adjustment for Bonferroni correction

Table 8 Comparing tumor features in Indian population with Caucasians

Features		Location of study				
		Our study from India (n = 321)	USA (Wills Eye hospital) (n = 7731)[7]	European Ophthalmic oncology group (n = 7369)[6]	Ophthalmic oncology task force (Worldwide multicenter) (n = 3217)[5]	Ocular oncology Centre, Liverpool, England (n = 3072)[17]
Distribution in T groups	T1/T2/T3/T4 (%)	11/23/39/22	46/27/21/6	24/33/31/12	35/35/24/6	NA
Mean age at presentation (years)	All cases (T1/T2/T3/T4)	47 (49/46/46/47)	58 (57/58/58/61)	NA	NA	62.8 (Median age at treatment)
Gender	Male %	58 (55/53/59/62)	51 (48/50/56/55)	NA	NA	51
	Female %	42 (44/47/41/38)	49 (52/50/44/45)	NA	NA	49
Quadrantic tumor location %	Superior	17 (11/15/23/12)	22 (25/21/20/18)	NA	NA	NA
	Nasal	17 (6/19/21/15)	21 (19/21/26/19)	NA	NA	NA
	Inferior	20 (22/27/18/13)	21 (19/22/23/19)	NA	NA	NA
	Temporal	34 (44/39/38/20)	33 (37/34/29/22)	NA	NA	NA
	Diffuse	12 (17/0/0/0/40)	3 (< 1/1/3/22)	NA	NA	NA
Anteroposterior tumor epicenter %	Macula	22 (61/25/18/8)	5 (10/2/1/< 1)	NA	NA	NA

Table 8 continued

Features		Location of study				
		Our study from India (n = 321)	USA (Wills Eye hospital) (n = 7731)[7]	European Ophthalmic oncology group (n = 7369)[6]	Ophthalmic oncology task force (Worldwide multicenter) (n = 3217)[5]	Ocular oncology Centre, Liverpool, England (n = 3072)[17]
	Macula to equator	42 (22/56/39/42)	73 (66/87/71/68)	NA	NA	NA
	Equator	12 (9/9/16/8)	16 (11/10/28/30)	NA	NA	NA
	Pre-equatorial	24 (9/10/27/38)	6 (13/< 1/< 1/2)	NA	NA	NA
Tumor color %	Pigmented	91 (78/94/89/97)	54 (54/48/54/70)	NA	NA	NA
	Amelanotic	7 (19/5/8/2)	15 (15/16/18/10)	NA	NA	NA
	Mixed	2 (3/1/3/1)	31 (31/36/27/21)	NA	NA	NA
Subretinal fluid %		89 (86/89/84/99)	73 (64/80/82/83)	NA	NA	NA
Extraocular extension %		5 (0/3/2/13)	2 (1/< 1/4/12)	NA	NA	NA
KM estimates at 5 years	Distant metastasis (%)	0/0/6.8/13	8/14/31/51	NA	3/15/23/39	NA
	Survival (%)	100/100/98/92	96/92/81/70	76-96/65-89/47/82/26-57	NA	NA

NA Data not available

follow-up duration, differences in tumor genetics, or unknown racial, socioeconomic and environmental factors.

The main limitations of this study include retrospective nature of the study, shorter follow-up duration, and lack of genetic testing. In conclusion, AJCC 8th edition classification on PUM in Asian Indian patients using the T subcategories is useful in predicting the prognosis. In an event of limited access to important prognostic tools like genetic analysis, such an anatomically based clinical classification is useful in patient counseling.

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Compliance with ethical standards

Conflict of interest All authors declare that he/she has no conflict of interest.

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards

Informed consent Informed consent was obtained from all individual participants or parents/guardians included in the study

References

1. Edge SB, Greene FL, Byrd DR et al (eds) (2017) AJCC cancer staging manual, 8th edn. Springer, New York
2. Edge SB, Byrd DR, Comptom CC et al (eds) (2010) AJCC cancer staging manual, 7th edn. Springer, New York
3. Finger PT (2009) 7th edition, AJCC-UICC ophthalmic oncology task force. The 7th edition AJCC staging system for eye cancer: an international language for ophthalmic oncology. Arch Pathol Lab Med 133(8):1197–1198
4. Kujala E, Kivelä T (2005) Tumor, node, metastasis classification of malignant ciliary body and choroidal melanoma evaluation of the 6th edition and future directions. Ophthalmology 112(6):1135–1144

5. AJCC Ophthalmic Oncology Task Force (2015) International validation of the American joint committee on cancer's 7th edition classification of uveal melanoma. JAMA Ophthalmol 133(4):376–383
6. Kujala E, Damato B, Coupland SE, Desjardins L, Bechrakis NE, Grange JD, Kivelä T (2013) Staging of ciliary body and choroidal melanomas based on anatomic extent. J Clin Oncol 31(22):2825–2831
7. Shields CL, Kaliki S, Furuta M, Fulco E, Alarcon C, Shields JA (2013) American joint committee on cancer classification of posterior uveal melanoma (tumor size category) predicts prognosis in 7731 patients. Ophthalmology 120(10):2066–2071
8. Shields CL, Kaliki S, Furuta M, Fulco E, Alarcon C, Shields JA (2015) American Joint Committee on Cancer Classification of Uveal Melanoma (Anatomic Stage) Predicts Prognosis in 7,731 Patients: the 2013 Zimmerman Lecture. Ophthalmology 122(6):1180–1186
9. Bagger M, Andersen MT, Andersen KK, Heegaard S, Andersen MK, Kiilgaard JF (2014) The prognostic effect of American joint committee on cancer staging and genetic status in patients with choroidal and ciliary body melanoma. Invest Ophthalmol Vis Sci 56(1):438–444
10. Kaliki S, Shields CL (2017) Uveal melanoma: relatively rare but deadly cancer. Eye (Lond) 31(2):241–257
11. Shields CL, Kaliki S, Furuta M, Mashayekhi A, Shields JA (2012) Clinical spectrum and prognosis of uveal melanoma based on age at presentation in 8,033 cases. Retina 32(7):1363–1372
12. Kashyap S, Venkatesh P, Sen S, Khanduja S, Shrey D, Tinwala S, Garg S (2014) Clinicopathologic characteristics of choroidal melanoma in a North Indian population: analysis of 10-year data. Int Ophthalmol 34(2):235–239
13. Dhupper M, Biswas J, Gopal L, Kumar SK, Khetan V (2012) Clinicopathological correlation of choroidal melanoma in Indian population: a study of 113 cases. Oman J Ophthalmol 5(1):42–45
14. Biswas J, Kabra S, Krishnakumar S, Shanmugam MP (2004) Clinical and histopathological characteristics of uveal melanoma in Asian Indians. A study of 103 patients. Indian J Ophthalmol 52(1):41–44
15. Pal BP, Garge S, Khetan V (2017) Choroidal melanoma: a short review with an Indian perspective. Oman J Ophthalmol 10(3):135–144
16. Meeralakshmi P, Shah PK, Narendran V (2017) Experiences of two different modalities in the management of choroidal melanoma in the Asian Indian population. South Asian J Cancer 6(3):134–136
17. Damato BE, Heimann H, Kalirai H, Coupland SE (2014) Age, survival predictors, and metastatic death in patients with choroidal melanoma: tentative evidence of a therapeutic effect on survival. JAMA Ophthalmol 132(5):605–613
18. Cormier JN, Xing Y, Ding M, Lee JE, Mansfield PF, Gershenwald JE, Ross MI, Du XL (2006) Ethnic differences among patients with cutaneous melanoma. Arch Intern Med 166(17):1907–1914
19. Shields CL, Kaliki S, Cohen MN, Shields PW, Furuta M, Shields JA (2015) Prognosis of uveal melanoma based on race in 8100 patients: the 2015 Doyne Lecture. Eye (Lond) 29(8):1027–1035